

Additional text 2: is kinase depletion by staurosporine explained by drug binding?

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Question

10 μ M staurosporine depletes protein kinases from both co-chaperone pull-downs (main report): the 346 kinases detected shift by -0.7 to -1.1 standard deviations, while the other ATP-binding proteins do not move. Some kinases, notably the PKC family, rise instead. If the depletion reflects the drug binding kinases and releasing them from the chaperone machinery, then (i) kinases that staurosporine does not bind should not be depleted, and (ii) tighter binders should be depleted more.

Data

Staurosporine dissociation constants (Kd) from ChEMBL for three kinome-wide competition-binding studies: Davis et al. 2011 (371 kinases), Karaman et al. 2008 (278) and Fabian et al. 2005 (102). Measurements on mutant or variant constructs (89 records) were excluded, and each kinase was given the median log10 Kd across studies. The studies agree closely (Spearman 0.98 Davis vs Karaman on 277 kinases, 0.95 and 0.93 for the comparisons with Fabian), giving 374 kinases with a consensus Kd (median 62 nM; 30 not bound at 10 μ M). 239-255 of the kinases in each GA_33 staurosporine target have a Kd.

The intended second test, against each kinase's dependence on the HSP90-CDC37 chaperone system (Taipale et al. 2012, *Cell*), could not be run: the supplementary table is served only behind a browser check, both on PubMed Central and on the MIT repository, so it cannot be fetched by script. It is the natural next test and is listed under next steps.

Results

1. Tighter binders are depleted more. Spearman correlation between log10 Kd and the staurosporine response (positive = tighter binding, stronger depletion):

Target	Kinases with Kd	rho	p	Binders, Kd < 1 μ M	Non-binders, Kd $\geq$ 10 μ M	Non-kinases
35 °C	239	+0.16	0.01	-0.21 (195)	+0.04 (18)	+0.00
37 °C	245	+0.14	0.03	-0.25 (201)	-0.04 (18)	+0.02
43 °C	254	+0.28	7e-6	-0.31 (208)	+0.01 (19)	+0.00

Mean log2 fold change, staurosporine vs control, the two co-chaperones averaged (number of kinases).

The kinases staurosporine does not bind behave like non-kinases ($p = 0.68, 0.07$ and 0.93 for the difference), unlike the binders. But there are only 18-19 of them, and the difference between binders and non-binders reaches significance only at 43 °C ($p = 0.07, 0.2$ and 0.02). The control points the right way without proving that binding is required.

2. The relation holds within kinase groups, and binding is graded in the cell. Correcting for kinase group (AGC, CAMK, CMGC, STE, TKL, tyrosine kinases, and others), the partial rank correlation of Kd with depletion is +0.12 [+0.00, +0.24] at 35 °C, +0.11 [-0.01, +0.23] at 37 °C and **+0.24 [+0.13, +0.34]** at 43 °C (bootstrap over kinases). At 10 μ M nearly every binder below 1 μ M would be saturated in a test tube, which predicts a step near micromolar; the data instead follow Kd across the nanomolar range. Occupancy computed with the test-tube Kd correlates more weakly with the response ($r = -0.15$ at 43 °C) than occupancy computed with Kd raised 1,000-fold ($r = -0.26$), as competition with cellular ATP would do.

3. The AGC exception. By Kd bin (37 °C), the tightest binders are depleted most, but the relation is broken by the AGC kinases; without them it falls off steadily with affinity:

Kd	< 1 nM	1-10 nM	10-100 nM	100 nM-1 μ M	1-10 μ M	$\geq 10 \mu$ M
All kinases	-0.71 (19)	-0.09 (43)	-0.25 (81)	-0.21 (58)	-0.13 (26)	-0.04 (18)
Without AGC kinases	-0.82 (16)	-0.24 (35)	-0.29 (66)	-0.22 (52)	-0.13 (26)	-0.04 (18)
AGC kinases	-0.12 (3)	+0.54 (8)	-0.07 (15)	-0.10 (6)		

The AGC kinases that rise sit mostly in the 1-10 nM bin:

AGC kinase	Kd (nM)	Response	AGC kinase	Kd (nM)	Response
PRKG1	32	+4.36	PRKCD	1.5	+0.37
PRKCQ	8.5	+2.24	PRKCA	50	+0.33
PRKCH	4.8	+1.85	PRKCE	0.2	+0.16
PKN2	2.5	+1.52	RPS6KB1	1.3	+0.12
PKN1	1.3	+1.24	AKT2	44	+0.00

These kinases bind staurosporine as tightly as any, yet gain co-chaperone association. Whatever staurosporine does to PKC, PKN and PKG, it is not release. Group by group, CAMK kinases fall most (mean -0.41, median Kd 16 nM) and AGC kinases rise (+0.07, median Kd 27 nM) at similar affinities.

4. Kinases most present in the control pull-downs lose the most. Among kinases, the level in the control pull-down predicts depletion (Spearman -0.22 to -0.34 with level and response taken from disjoint replicates, so not shared noise). In a joint rank model of the kinases:

Target	Kd	Level in control pull-down	R2: group	+ Kd	+ level
35 °C	+0.19 [+0.04, +0.32]	-0.39 [-0.51, -0.26]	3.9 %	5.3 %	18.8 %
37 °C	+0.17 [+0.02, +0.33]	-0.37 [-0.50, -0.25]	3.2 %	4.4 %	17.5 %
43 °C	+0.27 [+0.15, +0.40]	-0.32 [-0.44, -0.21]	9.7 %	14.8 %	25.0 %

Standardised rank coefficients with 95 % bootstrap intervals; R2 in sample.

Affinity and pull-down level contribute independently, and the level is the stronger. Across all proteins the level does not predict the staurosporine response ($\rho = -0.07$; 0.00 from disjoint replicates), and the kinase depletion is unchanged after adjusting for it (-0.60, -0.90 and -0.98 sd).

Interpretation

The kinase depletion behaves as drug-driven release from the co-chaperones would: it follows binding affinity, within kinase groups as well as between them, more strongly at 43 °C, along a graded curve that fits ATP competition in the cell better than a simple saturation model (not formally tested); and the few kinases staurosporine does not bind are not depleted, as far as 18 of them can show. Two further facts sharpen the picture. Kinases that are most present in the co-chaperone pull-downs to begin with lose the most, which is what release would predict

if that level reflects how strongly each kinase engages the chaperone machinery. And the AGC kinases with lipid- and nucleotide-sensing regulatory domains go the other way despite tight binding, which marks them as mechanistically different.

Critical evaluation

- **Binding data.** Three independent studies agree at Spearman 0.93-0.98, so the Kd values are not the weak point. They are measured on isolated kinase domains in vitro, not in cells.
- **The non-binder control is small.** 18-19 kinases; it is consistent with release but does not establish that binding is necessary.
- **Effect sizes.** Affinity explains a few percent of the variance among kinases (partial rho 0.11-0.24). Most of the kinase-to-kinase variation lies elsewhere, notably in the pull-down level and group.
- **What the pull-down level means.** It mixes how much of a kinase the cell contains with how strongly it engages the co-chaperone. Reading it as chaperone engagement is a hypothesis; the input lysate would separate the two (level in the pull-down relative to level in the lysate).
- **AGC kinases.** Their rise is clear but unexplained. One possibility to test: ATP-site inhibitors are known to change the regulatory state of some AGC kinases (for example by protecting activation-loop phosphorylation), which could alter their chaperone engagement in the opposite direction.
- **Not done.** The HSP90-CDC37 client-strength comparison (Taipale et al. 2012), because the data could not be downloaded by script.

Next steps

1. HSP90-CDC37 client strength (Taipale et al. 2012, *Cell* 150:987, Table S1): does it predict depletion beyond affinity and pull-down level?
2. The input lysate: depletion from the pull-down relative to depletion from the lysate.

Methods files

`code/ad02_kinase_affinity.py`; ChEMBL extracts in `data/external/kinase/`; results in `data/external/kinase/ad02_res`